PADHAKU
Science · Class 10 Science



CHAPTER 10

The Human Eye and the Colourful World

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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Questions · 10.2 Defects of Vision and their Correction

Q 1 **What is meant by power of accommodation of the eye?**

✔ ANSWER

The ability of the eye lens to **adjust its focal length** (with the help of the ciliary muscles) so that both near and distant objects are focused on the retina is called the **power of accommodation**.

Q 2 **A person with a myopic eye cannot see objects beyond 1.2 m distinctly. What should be the type of the corrective lens used to restore proper vision?**

✔ ANSWER

The person has **myopia** (far point = 1.2 m instead of infinity).

Corrective lens: a **concave (diverging) lens** of suitable power.

Its focal length = −1.2 m, so P = 1/f = 1/(−1.2) ≈ **−0.83 D**.

Q 3 **What is the far point and near point of the human eye with normal vision?**

✔ ANSWER

- **Near point:** about **25 cm** from the eye (least distance of distinct vision).

- **Far point: infinity**.

So a normal eye sees clearly objects between 25 cm and infinity.

Q 4 **A student has difficulty reading the blackboard while sitting in the last row. What could be the defect the child is suffering from? How can it be corrected?**

✔ ANSWER

The child cannot see distant objects clearly, so the defect is **myopia (near-sightedness)**.

It is corrected by using spectacles with a **concave lens** of suitable power, which brings the image back on to the retina.

⭐ Activities · 10.3 Refraction of Light through a Prism

Activity 10.1 **Trace a glass prism on paper, draw an incident ray PE on face AB, fix pins P, Q and find R, S through face AC. Draw the refracted ray EF, the emergent ray FS and the normals; mark ∠i, ∠r, ∠e and the angle of deviation ∠D.**

Compare the angle of incidence and angle of refraction at each surface. Is this similar to the bending in a glass slab? Mark and measure the angle of deviation.

✔ ANSWER

Observation: At face AB (air → glass) the ray bends **towards** the normal (∠r < ∠i). At face AC (glass → air) it bends **away** from the normal.

The bending at each surface is like that in a glass slab, but because the faces of a prism are **not parallel**, the emergent ray is **not parallel** to the incident ray. It bends towards the base of the prism.

The angle between the incident ray (produced) and the emergent ray is the **angle of deviation ∠D**; measure it with a protractor (its value depends on the prism and ∠i).

Conclusion: A prism deviates light because of refraction at two inclined surfaces.

⭐ Activities · 10.4 Dispersion of White Light by a Glass Prism

Activity 10.2

Let a narrow beam of sunlight from a slit in a cardboard fall on one face of a glass prism. Turn the prism until the light coming out falls on a screen.

What do you observe? Why does this happen? What is the sequence of colours on the screen?

✔ ANSWER

Observation: A beautiful band of seven colours (a **spectrum**) is seen on the screen.

Sequence: **Violet, Indigo, Blue, Green, Yellow, Orange, Red (VIBGYOR)**; red is nearest the top (least bent), violet nearest the base (most bent).

Reason: White light is made of seven colours. Different colours bend through different angles in the prism (red the least, violet the most), so they come out along different paths and separate.

Conclusion: The splitting of white light into its component colours is called **dispersion**.

📝 Exercises

1 **The human eye can focus on objects at different distances by adjusting the focal length of the eye lens. This is due to**
(a) presbyopia. (b) accommodation. (c) near-sightedness. (d) far-sightedness.

✔ ANSWER

(b) accommodation. The ability of the eye lens to adjust its focal length is called accommodation.

2 **The human eye forms the image of an object at its**
(a) cornea. (b) iris. (c) pupil. (d) retina.

✔ ANSWER

(d) retina. The eye lens forms a real, inverted image on the light-sensitive retina.

3 **The least distance of distinct vision for a young adult with normal vision is about**
(a) 25 m. (b) 2.5 cm. (c) 25 cm. (d) 2.5 m.

✔ ANSWER
(c) 25 cm. This is the near point of a normal eye.

4 **The change in focal length of an eye lens is caused by the action of the**
(a) pupil. (b) retina. (c) ciliary muscles. (d) iris.

✔ ANSWER
(c) ciliary muscles. They change the curvature, and so the focal length, of the eye lens.

5 **A person needs a lens of power −5.5 dioptres for correcting his distant vision. For correcting his near vision he needs a lens of power +1.5 dioptre. What is the focal length of the lens required for correcting (i) distant vision, and (ii) near vision?**

✔ ANSWER
Formula: f = 1/P (f in metres, P in dioptres).
(i) Distant vision: P = −5.5 D

f = 1/(−5.5) m = −0.182 m

f = −18.2 cm (concave lens)

(ii) Near vision: P = +1.5 D

f = 1/(+1.5) m = +0.667 m

f = +66.7 cm (convex lens)

6 **The far point of a myopic person is 80 cm in front of the eye. What is the nature and power of the lens required to correct the problem?**

✔ ANSWER

The lens must form the image of a distant object (u = ∞) at the far point, v = −80 cm.

1/f = 1/v − 1/u = 1/(−80) − 1/∞ = −1/80

f = −80 cm = −0.8 m

P = 1/f = 1/(−0.8) = **−1.25 D**

Nature: **concave (diverging) lens**.

7 **Make a diagram to show how hypermetropia is corrected. The near point of a hypermetropic eye is 1 m. What is the power of the lens required to correct this defect? Assume that the near point of the normal eye is 25 cm.**

✔ ANSWER

Diagram (draw as in Fig. 10.3): (a) Rays from an object at the normal near point N′ (25 cm) meet **behind the retina** in the hypermetropic eye. (b) A **convex lens** is placed in front of the eye: it bends the rays from N′ so they seem to come from the eye's near point N (1 m), and the eye then focuses them **on the retina**.

Calculation: Object at u = −25 cm; the lens must form its virtual image at the near point, v = −100 cm.

1/f = 1/v − 1/u = 1/(−100) − 1/(−25) = −1/100 + 4/100 = 3/100

f = 100/3 cm ≈ +33.3 cm = +0.333 m

P = 1/f = **+3.0 D** (convex lens)

8 **Why is a normal eye not able to see clearly the objects placed closer than 25 cm?**

✔ ANSWER

The focal length of the eye lens cannot be decreased below a certain **minimum limit**, even when the ciliary muscles contract fully.

So for objects closer than 25 cm the eye lens cannot focus the image on the retina; the image is blurred and the eye feels strain.

9 **What happens to the image distance in the eye when we increase the distance of an object from the eye?**

✔ ANSWER

The image distance **does not change**. It is the distance between the eye lens and the retina, which is fixed.

Instead, the eye lens becomes thinner and its **focal length increases** (accommodation), so the image is still formed on the retina.

10 **Why do stars twinkle?**

✔ ANSWER

Starlight entering the earth's atmosphere is **refracted continuously** by layers of air of changing refractive index (atmospheric refraction).

Since the atmosphere is not stationary, the path of the light keeps changing slightly, so the apparent position of the star fluctuates.

Stars are very far away and act as **point sources**, so the amount of light entering our eye **flickers**: the star looks brighter at times and fainter at others. This is twinkling.

11 **Explain why the planets do not twinkle.**

✔ ANSWER

Planets are much **closer** to the earth, so they are seen as **extended sources** (a collection of many point sources).

The changes in light from all these point sources **average out to zero**, so the total light entering the eye stays steady and planets do not twinkle.

12 **Why does the sky appear dark instead of blue to an astronaut?**

✔ ANSWER

The sky looks blue because fine particles of air **scatter** blue light of sunlight into our eyes.

At very high altitudes (outside the atmosphere) there is **no air to scatter** sunlight, so no scattered light reaches the astronaut's eyes and the sky appears **dark**.

🎉 **Chapter 10 done! Revise the answers once more tomorrow.**